

Question

M is an element of the sixth period. The flue gas obtained by leaching roasted molybdenite with water, the volatile oxide **N** of **M** (containing **M** 76.9%) is absorbed, and the resulting solution contains acid **A**; the solution is purified, concentrated, and alkaline gas **B** is passed into it, crystallization occurs, and **A** is converted into ammonium salt **C**; **C** is reduced by hydrogen at 1000°C to obtain **M** element. If **C** is directly heated to decompose, then oxide **D** is obtained.

Adding 6 mol/L hydrochloric acid to **A** and adding potassium iodide for reduction, a green paramagnetic potassium salt **E** can be prepared, its magnetic moment $\mu = 3.25 \mu_B$; adding AgNO_3 to the saturated solution of **E** yields an orange precipitate **F**; **F** decomposes upon heating to give silver chloride and two other chlorides **G**, **H**; **G**, **H** are both chlorine-bridged polymers, where **G** is a dimer and the metal is six-coordinate; **H** is a trimer, containing metal-metal bonds, and the ratio of bridging chlorine to terminal chlorine is 1 : 2. **G** can also be prepared by directly heating **M** element with chlorine gas to 400°C; **G** undergoes a disproportionation reaction with water to obtain a hydrate precipitate of **D** and a solution of **A** (the molar ratio of the products is 2 : 1).

The following statement is correct:

- A.** All other options are incorrect
- B.** Element **M** has an even number of unpaired electrons in its ground state.
- C.** Element **M** does not contain common radioactive nuclides.
- D.** The mass fraction of **M** in **D** is 79.5 %
- E.** The oxidation state of **M** in **E** is different from that of **D**.
- F.** The mass fraction of **M** in **F** is 34.2 %

G. The mass fraction of **M** in **G** is greater than **H**

H. The chemical equation for the reduction of **A** to obtain **E**, when the coefficients are balanced to the simplest integer ratio, has a sum of 37 for the coefficients of all substances.

Answer

Correct Answer: **H**

Detailed Explanation

The flue gas obtained from roasting molybdenite can yield the sixth-period **M**, indicating that the position of **M** on the periodic table is close to Mo, and it may be Ta, W, Re.

CHECKPOINT

1 PTS

The position of **M** on the periodic table is close to Mo

After roasting, elements are generally oxidized to their highest valence state. The volatile oxide **N** of **M** may be Ta_2O_5 , WO_3 , Re_2O_7 ; based on the mass fraction of 76.9%, **N** can be calculated to be Re_2O_7 . Therefore, element **M** is Re.

CHECKPOINT

1 PTS

N is Re_2O_7 , element **M** is Re

The valence electron configuration of Re is $5d^56s^2$, containing five unpaired electrons, so option B is incorrect. ^{187}Re is a radioactive nuclide, so option C is incorrect.

CHECKPOINT

1 PTS

^{187}Re is a radioactive nuclide

Re_2O_7 reacts with water to produce acid **A**, so **A** is HReO_4 ; its reaction with alkaline gas produces ammonium salt, so the ammonium salt **C** is obviously NH_4ReO_4 .

CHECKPOINT

1 PTS

Ammonium salt **C** is obviously NH_4ReO_4

NH_4ReO_4 decomposes upon direct heating. Re(VII) in the molecule has strong oxidizing properties, which can oxidize the ammonium ion to nitrogen gas, while it is reduced to the stable valence oxide ReO_2 ; that is, **D** is ReO_2 .

CHECKPOINT

1 PTS

Re(VII) has strong oxidizing properties, which can oxidize the ammonium ion to nitrogen gas, while it is reduced to the stable valence oxide ReO_2

CHECKPOINT

1 PTS

D is ReO_2

HReO_4 is reduced by hydrochloric acid and potassium iodide to obtain potassium salt **E**. According to the magnetic moment $\mu = 3.25 \mu_B = \sqrt{n(n+2)} \mu_B$, it can be calculated that **E** contains 3 unpaired electrons. (The theoretical magnetic moment should be $\mu = 3.87 \mu_B$, but the sixth-period elements have a large deviation due to relativistic effects.)

CHECKPOINT

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According to the magnetic moment $\mu = 3.25 \mu_B = \sqrt{n(n+2)} \mu_B$, it can be calculated that **E** contains 3 unpaired electrons

Therefore, **E** contains Re(IV) ; considering the stable octahedral hexacoordination, the chemical formula of **E** is K_2ReCl_6 .

CHECKPOINT

1 PTS

E contains Re(IV)

CHECKPOINT

1 PTS

E chemical formula is K_2ReCl_6

Adding nitrate to K_2ReCl_6 yields a precipitate. According to the principle of like dissolves like, the precipitate **F** is Ag_2ReCl_6 .

CHECKPOINT

1 PTS

According to the principle of like dissolves like, the precipitate **F** is Ag_2ReCl_6

According to the chemical formulas, options D, E, and F are all incorrect.

G contains chlorine bridges and is dimeric, which is obviously $\text{Re}_2\text{Cl}_{10}$ with two octahedra connected by edge-sharing. **H** is trinuclear and contains metal-metal bonds, and should be a triangular configuration; there should be 3 bridging chlorines as the sides of the triangle, and the terminal chlorines are 6 according to the ratio, so **H** is Re_3Cl_9 . From the chemical formula, option G is incorrect.

CHECKPOINT

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G is $\text{Re}_2\text{Cl}_{10}$ with two octahedra connected by edge-sharing

CHECKPOINT

1 PTS

H is Re_3Cl_9

HReO_4 is reduced by hydrochloric acid and potassium iodide to obtain potassium salt K_2ReCl_6 , and iodine is oxidized to elemental form. Therefore, the balanced equation is:



CHECKPOINT

1 PTS



The sum of the coefficients is 37, so option H is correct.